
Trajectory-Aware Spiking DiTs Conversion via Membrane Potential Error-Feedback

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Abstract

Diffusion Transformers (DiTs) have achieved state-of-the-art generative performance, yet their iterative denoising process remains computationally expensive and energy-intensive. Spiking Neural Networks (SNNs) offer a promising neuromorphic alternative for energy efficiency; however, the non-differentiable nature of spiking neurons makes direct training difficult, positioning ANN-to-SNN conversion as a more practical, training-free solution. In this paper, we identify a critical challenge unique to converting DiTs: standard fixed-scale spiking neurons fail to accommodate the highly dynamic activation ranges inherent across denoising steps. This mismatch leads to cumulative errors that significantly degrade generation fidelity. To resolve this, we propose a novel conversion framework featuring Multi-Threshold (MT) neurons and a Membrane Potential Error-Feedback (MPEF) mechanism. MT neurons expand the expressive capacity of discrete spikes by employing a multi-level firing strategy. Concurrently, MPEF exploits the temporal correlation between successive denoising steps to recycle residual membrane potential, effectively compensating for information loss and mitigating distribution shifts without re-training. Extensive experiments on ImageNet demonstrate that our framework achieves competitive generative quality with superior energy efficiency, establishing a new performance benchmark for spiking Diffusion Transformers. Our code is available at <https://github.com/JustVelkhana/SpikingDiT>.

1. Introduction

Diffusion Transformers (DiTs) (Peebles & Xie, 2023) have emerged as a powerful class of generative models, achieving exceptional performance in high-fidelity image synthesis by replacing traditional U-Net (Ronneberger et al., 2015) backbones with scalable Transformer architectures. By leveraging the representation learning capabilities of Transformers, DiTs exhibit superior scalability and state-of-the-art results across various benchmarks, a paradigm whose profound potential is further exemplified by the success of Sora (Brooks et al., 2024) as a large-scale DiT-based video generation model. However, the iterative nature of the denoising process requires dozens or even hundreds of denoising steps, each involving intensive floating-point computations within the massive Transformer layers. This repetitive execution leads to prohibitive computational overhead and high energy consumption, significantly hindering the deployment of large-scale DiTs on resource-constrained edge devices.

Spiking Neural Networks (SNNs) replicate the information processing mechanisms found in the brains of living organisms, offering enhanced biological interpretability, event-driven properties, and reduced power consumption (Ghosh-Dastidar & Adeli, 2009; Tavanaei et al., 2019). In contrast to conventional Artificial Neural Networks (ANNs), which process continuous-valued activation information synchronously, SNNs encode all information in discrete spike sequences (Eshraghian et al., 2023; Kim et al., 2022). This enables SNNs to perform cumulative operations with lower power budgets, thereby improving energy-efficiency. Therefore, the spike-based computing paradigm offers a highly promising solution for deploying DiTs in resource-constrained scenarios.

Recent endeavors have sought to bridge this gap by implementing diffusion models within the spiking domain, yet they predominantly rely on direct training. Representative works, such as SDDPM (Cao et al., 2024) and Spiking-Diffusion (Liu et al., 2024), are predicated on convolutional U-Net backbones and are optimized via direct training with surrogate gradients (Neftci et al., 2019). More recently, SDiT (Yang et al., 2024) has explored the integration of DiTs by leveraging the RWKV mechanism (Peng et al., 2023) to facilitate more efficient training. Despite these

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architectural innovations, these approaches primarily rely on training from scratch, which remains computationally expensive and often limits the models to small-scale regimes with restricted parameter counts.

Alternatively, ANN-to-SNN conversion (Diehl et al., 2015) provides a training-free approach to directly convert pre-trained ANN models into their spiking versions without the cost of retraining. Through precise weight mapping and neuron conversion, this paradigm preserves the expressive power of the original ANNs while significantly improving energy-efficiency (Bu et al., 2023). The validity of these conversion techniques has been validated across multiple domains, with recent works such as SpikeLLM (Xing et al., 2024) successfully scaling SNNs to the complexity of Large Language Models (LLMs). Nevertheless, the transition from single-pass LLMs inference to the repetitive denoising process of DiTs presents a unique challenge. Specifically, conventional conversion methods rely on fixed-scale spiking representations that fail to accommodate the dynamic activation evolving throughout the denoising trajectory, leading to significant representation errors and degraded generation quality.

To overcome the challenges arising from both the Transformer architecture and the denoising process, we propose a training-free ANN-to-SNN conversion framework specifically designed for DiTs. We employ Multi-Threshold (MT) neurons (Huang et al., 2024; 2025) to enrich the diversity of fixed-scale spiking representations. Crucially, to tackle the most prominent challenge unique to DiTs—where time-activation distributions continuously shift along the denoising trajectory—we introduce a membrane potential error-feedback (MPEF) mechanism. Leveraging the inherent similarity of activation distributions between adjacent denoising steps, MPEF utilizes the residual membrane potential as an error-feedback estimator. By carrying these residuals over as the initial state for the subsequent step, the mechanism implicitly mitigates the adverse effects of shifting activation distributions.

The main contributions of this work are as follows:

- We present the first training-free framework to bridge the gap between DiTs and energy-efficient SNNs. This framework integrates a suite of synergistic techniques, including MT neurons and the MPEF mechanism.
- We introduce MPEF, a novel mechanism that addresses time-varying activation shifts by carrying over the residual membrane potential to initialize the subsequent step. This approach effectively mitigates conversion-induced discrepancies and enhances the representational fidelity of spiking DiTs.
- Extensive evaluations on ImageNet demonstrate that

our framework achieves competitive performance, setting a new state-of-the-art for conversion-based Spiking DiTs.

2. Related work

2.1. Diffusion Transformers

Diffusion models based on UNet have made significant advancements in image generation (Ho & Salimans, 2022; Rombach et al., 2022; Saharia et al., 2022). However, their reliance on convolutional structures with localized receptive fields limits their ability to capture long-range dependencies and global semantic relationships, resulting in reduced performance on complex scenes and high-dimensional tasks (Peebles & Xie, 2023). In contrast, DiTs, by incorporating the self-attention mechanism, enable global modeling of feature dependencies, leading to superior performance in image generation tasks (Esser et al., 2024). This architectural shift has catalyzed various state-of-the-art models, ranging from the pioneering DiT (Peebles & Xie, 2023) and the efficient PixArt- α (Chen et al., 2023) to the large-scale video foundation model Sora (Brooks et al., 2024). However, the high computational overhead of DiTs, exacerbated by iterative denoising and the quadratic complexity of attention, motivates the exploration of energy-efficient alternatives such as SNNs.

2.2. ANN-to-SNN Conversion

ANN-to-SNN conversion has gained widespread recognition and adoption in recent years. By repurposing the weights of pre-trained ANNs, SNNs can achieve competitive performance without the need for additional training (Rueckauer et al., 2017; Bu et al., 2023). Mainstream ANN-to-SNN conversion methods are primarily rooted in rate coding, which correlates ReLU activations with the firing rates of Integrate-and-Fire (IF) neurons (Cao et al., 2015). Recent studies have explored advanced ANN-to-SNN conversion methodologies, extending their applicability to various activation functions and Transformer-based architectures. (Bu et al., 2023) proposes the QCFS activation function as a replacement for the ReLU activation function in source ANNs, which can better approximate the activation function of SNNs. (You et al., 2024) addresses the operator equivalence challenge by introducing a novel spiking equivalence self-attention. (Xing et al., 2024) proposes a spiking framework as a general alternative to traditional quantizers, scaling up SNNs to LLMs.

Despite these advancements, approximation errors persist in ANN-to-SNN conversion. These discrepancies are often quantified as residual membrane potential (Han et al., 2020; Hao et al., 2023b) and can be categorized into quantization, clipping, and unevenness errors (Hao et al., 2023a).

Critically, these errors are magnified in DiTs due to their iterative nature. Unlike previous architectures, the temporal evolution of activation distributions in DiTs challenges the fixed-scale spikes, leading to a divergent generative trajectory.

3. Preliminary

3.1. Diffusion Process

The foundation of diffusion models lies in a forward diffusion process, which is defined as a Markov chain that gradually adds Gaussian noise to the data (Ho et al., 2020). Given a data distribution $x_0 \sim q(x)$, the forward process $q(x_{1:N}|x_0)$ is constructed by iteratively injecting noise according to a pre-defined variance schedule $\beta_1, \dots, \beta_N$:

$$(x_\tau|x_{\tau-1}) = \mathcal{N}(x_\tau; \sqrt{1 - \beta_\tau}x_{\tau-1}, \beta_\tau \mathbf{I}) \quad (1)$$

As $\tau \rightarrow N$, the original data x_0 is iteratively corrupted until it becomes indistinguishable from Gaussian noise $\mathcal{N}(0, \mathbf{I})$, providing a stochastic latent space for the generative model (Sohl-Dickstein et al., 2015).

Remark 3.1. To avoid ambiguity, we distinguish between two types of temporal dimensions throughout this paper:

- **Diffusion Timesteps (τ):** We use $\tau \in \{1, \dots, N\}$ to denote the global denoising trajectory of DiTs, where N is the total number of sampling steps.
- **SNNs Simulation Steps (t):** We use $t \in \{1, \dots, T\}$ to represent the internal temporal dynamics of spiking neurons within a single denoising step, where T is the simulation time window.

3.2. Direct Coding ANN-to-SNN Conversion

In the traditional ANN-to-SNN conversion framework, the firing behavior of the IF neuron is commonly characterized by its firing rate, or equivalently, by the average postsynaptic potential (Bu et al., 2023). This relationship can be mathematically formulated as:

$$\psi[T] = \sum_{t=1}^T s[t] \cdot \lambda \quad (2)$$

where $\psi[T]$ represents the SNNs activation level within the T latency. The binary spikes $s[t]$ represent the spiking states at time step t , where a value of 1 indicates a firing event. The scalar λ corresponds to the firing threshold. This allows $\psi[T]$ in SNNs to approximate activations a in ANNs.

Conversion error. Such an approximation inherently introduces significant conversion errors. Similar to (Hao et al., 2023b), such conversion errors are quantified by the residual

membrane potential $v[T]$ at the final simulation step T as follows:

$$a = v[T] + \sum_{t=1}^T s[t] \cdot \lambda \quad (3)$$

4. Observations and Motivations

While DiTs demonstrate powerful generative capabilities, their deployment on neuromorphic hardware is hindered by the fundamental mismatch between the continuous, high-dynamic-range activations of DiTs and the discrete, fixed-scale representation of SNNs. Recent studies on quantization (Wu et al., 2024; Liu & Zhang, 2024; Zhao et al., 2024; Chen et al., 2025) imply that this difficulty stems from the complex distribution characteristics of DiTs. In this section, we analyze these distribution characteristics from both spatial and temporal perspectives to identify the root causes of conversion errors and motivate our proposed solutions.

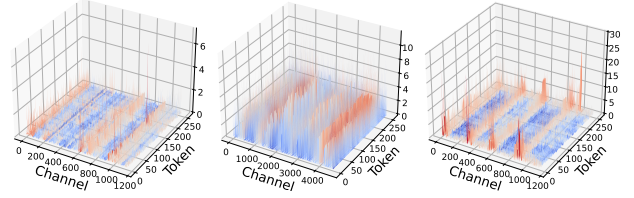


Figure 1. Distributions of activations in different modules of DiT-XL-2-256. The colormap is scaled such that warmer hues (red) represent peak activation intensities, while cooler hues (blue) correspond to regions of lower activation.

Observation 1: Spatial Outliers and Channel-wise Imbalance. First, we examine the spatial variants of activations within each module. As illustrated in Figure 1, DiTs exhibit severe channel-wise imbalance. A small subset of channels contains extreme outliers (shown in red) that are orders of magnitude larger than the majority. This characteristic poses a dilemma for ANN-to-SNN conversion: setting a firing threshold to accommodate these outliers causes the majority of small-value activations to fail to fire, while setting a lower threshold leads to severe clipping of the salient features.

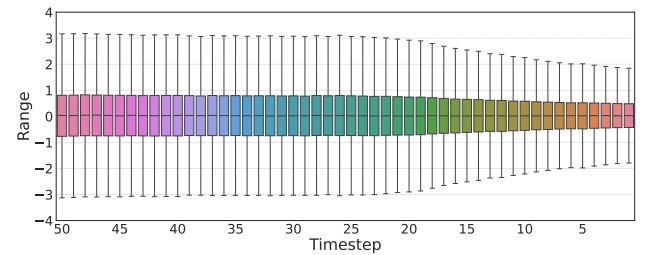


Figure 2. Timestep-wise activation variants of DiTs. Boxplots show the range and density of activations over 50 denoising steps.

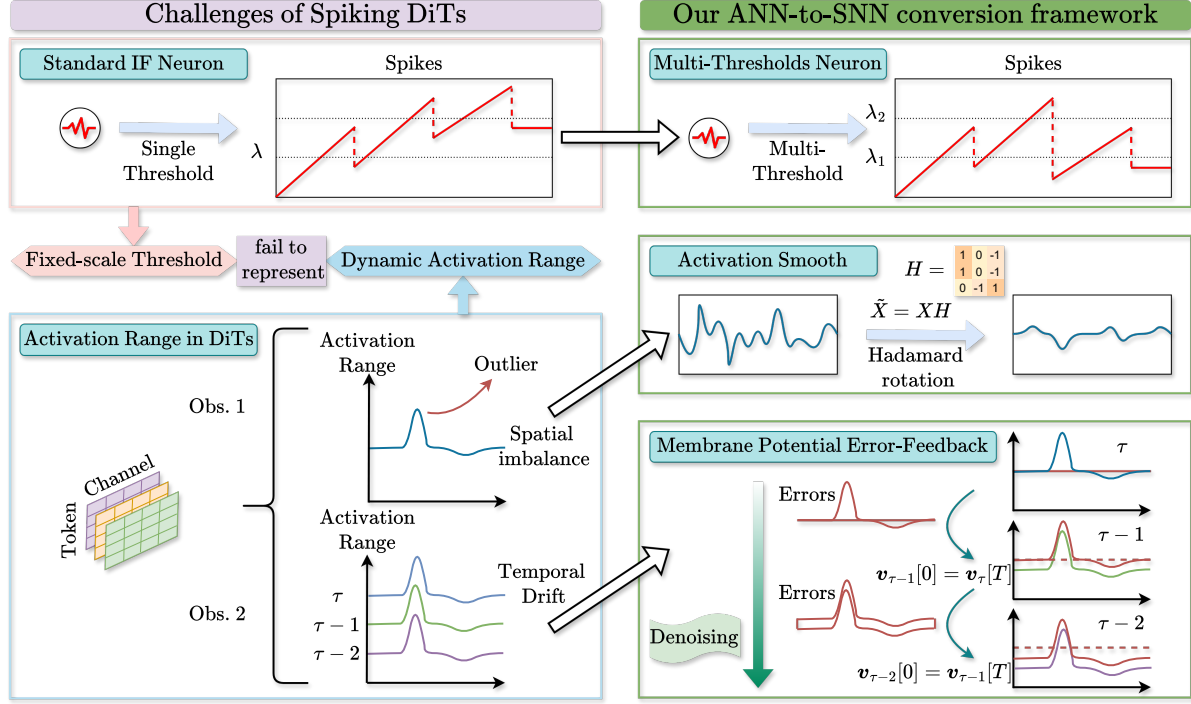


Figure 3. The overview of the proposed Spiking DiTs framework. The left illustrates the primary challenges of spiking DiTs: the mismatch between fixed-scale thresholds and dynamic activation ranges. The right presents our ANN-to-SNN conversion framework, featuring MT Neuron, activation smooth and MPEF mechanism.

Observation 2: Temporal Range Drifts. The challenge is further exacerbated by the iterative nature of the diffusion process. Unlike other models with static inputs, the activation ranges in DiTs undergo dynamic evolution as the noise is progressively removed. As shown in Figure 2, the statistical range of activations shifts significantly across the denoising trajectory.

This implies that a static conversion configuration derived from a single step is inherently suboptimal for the entire generation process, leading to accumulated errors along the denoising trajectory.

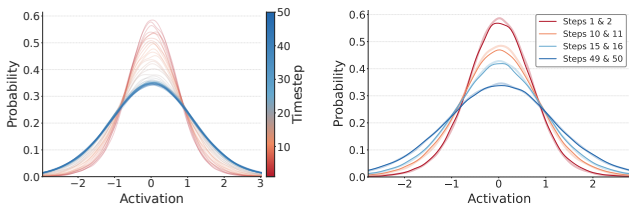


Figure 4. Probability density distributions of activations in DiTs. (Left) The evolution of activation distributions across the entire denoising trajectory (from 50 to 1). (Right) A comparison of activation distributions between adjacent denoising steps, showing the near-perfect overlap between consecutive steps.

Observation 3: Temporal Distribution Similarity. While the global range drifts drastically (Obs. 2), our analysis reveals a critical local property that can be exploited. As visualized in Figure 4, the probability densities of activations between adjacent denoising steps (τ and $\tau-1$) exhibit substantial overlap. This temporal consistency indicates that the conversion error is not random noise but a structured signal that evolves gradually.

Motivation: The interplay between spatial imbalance and temporal drifts creates a fundamental spatiotemporal discrepancy that hinders the effective ANN-to-SNN conversion of DiTs. As shown in the left panel of 3, standard fixed-scale neurons struggle to represent dynamic activation range, forcing an inevitable trade-off between representational precision and numerical clipping. However, the discovery of trajectory-consistency similarity (Obs. 3) offers a promising perspective: if the conversion residuals are temporally correlated, they can potentially be compensated for dynamically.

5. Method

Guided by the necessity to resolve the **spatiotemporal discrepancies** identified in Section 4, we propose a new ANN-to-SNN conversion framework for Spiking DiTs, as illus-

trated in Figure 3. Our framework is designed to bridge the representation gap by addressing conflict between fixed-scale spiking thresholds and the dynamic activation ranges inherent in the diffusion process. Specifically, we first perform activation smoothing to homogenize the spatial distribution. To further enhance the representation granularity within low latency, we replace standard binary IF neurons with MT neurons, significantly increasing the information capacity per time step. Finally, we introduce the MPEF mechanism to stabilize the denoising trajectory. By recycling the residual membrane potential from the current denoising step to initialize the neurons in the next step, MPEF establishes a temporal error-feedback loop that leverages the distributional similarity between adjacent stages, preventing the SNNs’ generative path from diverging from the original ANNs’ trajectory.

Activation Smoothing via Hadamard Rotation. As established in Obs.1, the extreme spatial imbalance in DiTs create a bottleneck for fixed-scale spiking thresholds. To mitigate this, we adopt Hadamard rotation (Ashkboos et al., 2024; Liu & Zhang, 2024) as a foundational pre-processing step. Specifically, we apply an orthogonal transformation $\tilde{\mathbf{X}} = \mathbf{X}\mathbf{H}$ to the activations $\mathbf{X}$. This operation effectively redistributes the concentrated outliers across all channels and transforms the imbalanced distribution into a more uniform and bounded distribution $\tilde{\mathbf{X}}$, thereby significantly alleviating the conversion error and providing a stable starting point for the subsequent processing.

5.1. Multi-Threshold Neurons

The replacement of continuous activation values with discrete outputs from spiking neurons has proven to be a challenge in ANN-to-SNN conversion, resulting in many previous works using various methods to overcome it (You et al., 2024). To enhance the expressivity of our spiking counterpart while maintaining low latency, we incorporate MT neurons (Huang et al., 2024; 2025). Unlike standard IF neurons that are restricted to binary states $\{0, 1\}$, the MT neuron achieves higher numerical granularity by employing a set of n thresholds, allowing it to emit multi-intensity spikes. Specifically, the firing logic of an MT neuron at simulation step t can be formulated as:

$$\mathbf{m}[t] = \mathbf{v}[t-1] + \mathbf{X}[t], \quad (4)$$

$$\mathbf{s}_i[t] = \text{MT}_{\theta,n}(\mathbf{m}[t]), \quad (5)$$

$$o[t] = \mathbf{s}_i[t]\lambda_i, \quad (6)$$

$$\mathbf{v}[t] = \mathbf{m}[t] - o[t], \quad (7)$$

$$\text{MT}_{\theta,n}(\mathbf{m}[t]) = \begin{cases} 0, & \text{if } \lambda_{2n} < \mathbf{m}[t] < \lambda_n \\ 1, & \text{elif } i = \arg \min_p |\mathbf{m}[t] - \lambda_p| \end{cases} \quad (8)$$

where there is a B -fold relationship between multiple thresholds and the corresponding negative thresholds as follows:

$$\theta = \lambda_1 = B\lambda_2 = \dots = B^{n-1}\lambda_n, \quad (9)$$

$$-\theta = \lambda_{n+1} = B\lambda_{n+2} = \dots = B^{n-1}\lambda_{2n} \quad (10)$$

To improve hardware feasibility, we set $B = 2$ in our experiments, enabling multiplication and division by 2 to be realized through bit shifting in actual deployment. Based on the MT neuron, the continuous activation $\mathbf{a}$ of the original ANNs is encoded as the summation of the spiking outputs over T time steps, which can be expressed as follows:

$$\mathbf{a} = \sum_{t=1}^T o[t] + \mathbf{v}[T] \quad (11)$$

where $\mathbf{v}[T]$ denotes the residual membrane potential at the final time step. Mathematically, this residual represents the error between the ANNs and SNNs.

In the specific case of DiTs, this residual error becomes particularly substantial due to channel-wise imbalance and temporal variations along the denoising trajectory. Despite its importance, this residual is typically ignored because the inference is treated as a complete and independent process in vanilla SNNs. However, DiTs operate through a sequence of iterative denoising steps, meaning the model does not stop after one pass. Given the temporal distribution similarity in DiTs, this residual is no longer regarded as a waste, but rather as a valuable prior that can be carried forward to compensate for the representation error in subsequent steps.

5.2. Membrane Potential Error-Feedback

Conventional SNNs typically initialize membrane potential to zero at the onset of an inference cycle. When applied to the iterative denoising process of DiTs, this standard practice effectively treats each denoising step as an isolated event, enforcing a complete reset of membrane potentials between independent steps. However, this isolation overlooks the continuous nature of the diffusion trajectory. The residual membrane potential $\mathbf{v}_\tau[T]$ remaining at the end of a step mathematically represents the conversion error. By discarding this residual, the spiking DiTs suffer from error accumulation, causing a drift from the ANNs trajectory.

To mitigate this representation error, we propose the Membrane Potential Error-Feedback (MPEF) mechanism. This approach exploits the temporal distribution similarity inherent in diffusion models, where activations evolve smoothly between adjacent steps. Unlike vanilla SNNs that enforce a “hard reset” of the membrane potential between independent inference steps, MPEF recycles the residual membrane potential $\mathbf{v}_\tau[T]$ from the current step and leverages it to initialize the spiking neurons for the subsequent denoising step. Specifically, the initial membrane potential for the

next denoising step $\tau - 1$ is inherited from the final state of step τ :

$$v_{\tau-1}[0] = v_\tau[T] \quad (12)$$

where τ and T denote the indices of the denoising steps and SNNs time-step, respectively. While $v[0] = 0$ in vanilla SNNs, MPEF carries this residual forward to the next step.

In order to visually demonstrate the integration of MPEF in the inference process, we have provided a complete denoising process in Algorithm 1.

Algorithm 1 Spiking DiTs Denoising Process with MPEF

Require: Noisy latent $\mathbf{z}_N$, Denoising steps N , SNN simulation time T , Initialize membrane potentials for all spike neurons: $v_{N+1}[T] \leftarrow 0$

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1: for  $\tau = N$  down to 1 do
2:   MPEF:  $v_\tau[0] \leftarrow v_{\tau+1}[T]$ 
3:   for  $t = 1$  to  $T$  do
4:     Compute forward pass for all layers
5:     Generate spikes  $\mathbf{s}_\tau[t]$  and update membrane potential  $v_\tau[t]$ 
6:   end for
7:    $\mathbf{z}_{\tau-1} \leftarrow \text{Denoising}(\mathbf{z}_\tau)$  {Standard DiTs update}
8: end for
9: return  $\mathbf{z}_0$ 
    
```

Several works (Yao et al., 2024; Li & Du, 2025) have also leveraged the temporal correlation of errors across adjacent denoising steps of diffusion models, to mitigate progressive error accumulation. However, our MPEF and these approaches differ fundamentally in their problem formulation and error utilization mechanisms. Existing methods primarily target quantization errors and explicitly construct correction terms (e.g., via distillation learning or analytical correction) to adjust model outputs. In contrast, our work addresses residual membrane potential error in the context of ANN-to-SNN conversion. We do not introduce additional correction modules or learning processes; instead, we allow errors to propagate between timesteps as part of the system state, forming an implicit compensation mechanism based on dynamic feedback.

Functionally, MPEF operates as a temporal negative feedback loop. If a spiking neuron at step τ residue a positive membrane potential $v_\tau[T] > 0$, this retained potential is carried forward, effectively lowering the firing threshold for step $\tau - 1$. This forces the spiking neuron to fire earlier or more frequently in the next step, thereby dynamically compensating for the previous error. By bridging the gap between discrete denoising steps, MPEF transforms the denoising process into a dynamic error-feedback system.

To theoretically justify this mechanism, we provide a formal analysis of the error mitigation capability of MPEF.

Consider a spiking neuron receiving input activation x_τ at diffusion step τ . Let the neuron produce an output y_τ , and let $V_\tau[T]$ denote the residual membrane potential after firing. With MPEF, the neuron dynamics can be written as:

$$V_\tau[T] = x_\tau + V_{\tau+1}[T] - y_\tau, \quad V_{N+1}[T] = 0 \quad (13)$$

Rearranging the equation and summing the above equation over $\tau = N, \dots, 1$, we obtain:

$$\sum_{\tau=1}^N y_\tau = \sum_{\tau=1}^N x_\tau + V_{N+1}[T] - V_1[T] \quad (14)$$

Since $V_{N+1}[T] = 0$, this simplifies to $\sum_{\tau=1}^N (x_\tau - y_\tau) = V_1[T]$. Due to the firing mechanism of spiking neurons, the residual membrane potential is confined to a bounded interval:

$$|V_\tau[T]| \leq \delta, \quad \forall \tau \quad (15)$$

where δ depends on the neuron’s firing threshold and discretization granularity. Then from the telescoping identity, we obtain:

$$\left| \sum_{\tau=1}^N (x_\tau - y_\tau) \right| = |V_1[T]| \leq \delta \quad (16)$$

This leads to the following result: Under MPEF, the accumulated discrepancy between input activations and spike-based outputs over the trajectory is uniformly bounded by a constant δ .

While in standard ANN-to-SNN conversion without MPEF, the neuron dynamics can be written as:

$$y_\tau = x_\tau - V_\tau[T] \quad (17)$$

This leads to a different result:

$$\left| \sum_{\tau=1}^N (x_\tau - y_\tau) \right| = \left| \sum_{\tau=1}^N V_\tau[T] \right| \leq N\delta \quad (18)$$

This indicates that MPEF confines the error within a bounded interval, whereas the error from standard ANN-to-SNN conversion may accumulate over the diffusion steps N .

Now we extend the analysis to the expected error of spiking conversion. For a layer of spiking neurons, let $\mathbf{a}_\tau \in \mathbb{R}^d$ be the ANN activation at denoising step τ , and $\sum_{t=1}^T \mathbf{o}_\tau[t]$ the corresponding SNN output. In the standard conversion without MPEF, the residual is discarded, and the expected error is

$$\mathcal{E}_\tau = \mathbb{E} \left[\left\| \mathbf{a}_\tau - \sum_{t=1}^T \mathbf{o}_\tau[t] \right\| \right] = \mathbb{E} [\|v_\tau[T]\|] \quad (19)$$

where $v_\tau[T]$ is the residual membrane potential after T time steps. Under MPEF, the conservation at step τ reads

$$a_\tau + v_{\tau+1}[T] = \sum_{t=1}^T o_\tau^{\text{MPEF}}[t] + v_\tau^{\text{MPEF}}[T] \quad (20)$$

Based on Obs.3, DiTs exhibit a smooth temporal evolution of the activation distribution, i.e., $P(a_\tau) \approx P(a_{\tau+1})$. With fixed firing thresholds, statistically similar inputs tend to produce highly correlated output spike patterns; consequently, the residual membrane potentials of adjacent steps are strongly coupled in the sense that their difference is typically much smaller than the residual magnitude itself:

$$\mathbb{E}[\|v_{\tau+1}[T] - v_\tau[T]\|] \ll \mathbb{E}[\|v_\tau[T]\|]. \quad (21)$$

In the MPEF framework, the injected residual $v_{\tau+1}[T]$ acts as an error-feedback term that compensates the discrepancy, driving $v_\tau^{\text{new}}[T]$ toward $v_{\tau+1}[T]$. Consequently, the instantaneous error

$$\mathcal{E}_\tau^{\text{new}} = \mathbb{E}[\|v_\tau^{\text{new}}[T] - v_{\tau+1}[T]\|] \quad (22)$$

is expected to be substantially smaller than the no-feedback error $\mathcal{E}_\tau = \mathbb{E}[\|v_\tau[T]\|]$, consistent with the bounded global accumulation derived above.

6. Experiments

In this section, we evaluate the effectiveness of our proposed Spiking DiTs with MT neurons and MPEF mechanism.

6.1. Experimental Setup

Models and Datasets. We perform experiments on the standard class-conditional ImageNet dataset (256×256 and 512×512 resolutions). Our primary backbone architecture is DiT-XL/2.

Implementation Details. We implement our spiking conversion framework using PyTorch. The SNN simulation timestep is set to a low-latency setting of $T = 4$. For the multi-threshold mechanism, we experiment with threshold counts $n \in \{6, 8, 10\}$, which correspond to an information capacity comparable to roughly 8-bit activations. For the diffusion sampling process, we utilize the steps of $N \in \{25, 50, 100\}$ to evaluate performance under both accelerated and standard generation scenarios. we report results using two baseline Classifier-Free Guidance (CFG) scales: $\text{cfg} = 1.5$ (Text-guided Image Generation) and $\text{cfg} = 1.0$ (Unconditional Generation). All experiments are conducted on NVIDIA 4090 GPUs.

Baselines and Comparisons. We compare our method with state-of-the-art post-training quantization (PTQ) methods for diffusion models: Q-Diffusion (Li et al., 2023),

PTQ4DiT (Wu et al., 2024), and Q-DiT (Chen et al., 2025). These methods are evaluated under W4A8 (4-bit weights/8-bit activations) and W8A8 settings. Our goal is to demonstrate that our SNN-based approach can achieve competitive or superior generation quality compared to these ANNs quantization methods while benefiting from the energy efficiency of spike-based accumulation.

Evaluation Metrics. To provide a comprehensive assessment of the generated image quality, we report the following metrics evaluated on 10,000 generated samples:

- Fréchet Inception Distance (FID) (Heusel et al., 2017): The primary metric for assessing visual quality and diversity.
- Spatial FID (sFID) (Szegedy et al., 2016): A variant of FID that is more sensitive to spatial structure and high-level artifacts.
- Inception Score (IS) (Salimans et al., 2016): Measures the clarity and diversity of generated images.
- Precision (Sajjadi et al., 2018): Evaluated to decouple the fidelity of the generative distribution.

6.2. Main Result

Table 1 and Table 2 present the quantitative comparison between our proposed Spiking DiTs and state-of-the-art PTQ methods. Our framework achieves near-lossless generative quality across various configurations. For DiT-XL/2 at 256×256 resolution, our method attains an FID of 5.86 using $N = 50$ denoising steps with only $T = 4$ simulation steps and $n = 6$ thresholds under $\text{cfg} = 1.5$. Visual results in Figure 5 corroborate our quantitative findings. Naive conversion methods often produce images with color shifting or fragmented structures when forced into low-latency spiking regimes. In contrast, our method produces high-fidelity samples that are visually indistinguishable from the ANNs baseline.

6.3. Ablation Study

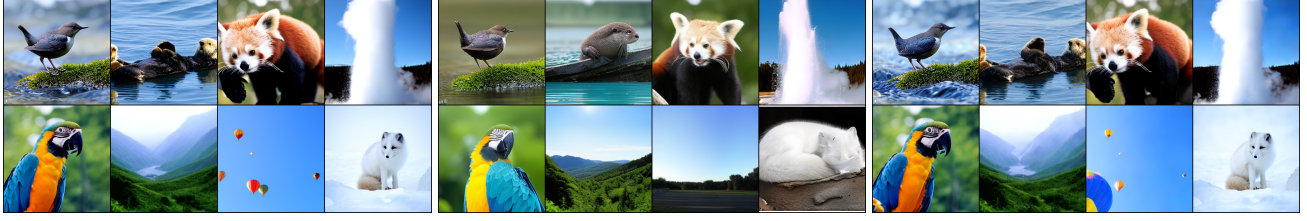
To verify the individual contribution of MPEF mechanism, we conduct ablation experiments on the DiT-XL-2-256 with $T = 4$ and $n = 6$. We compare these configurations:

Naive Conversion: The SNN model is converted from the pre-trained DiTs ANN model.

Conversion + Hadamard rotation: Hadamard rotation helps redistribute activation and improve the performance of ANN to SNN conversion.

Conversion + MPEF: The residual membrane potential from step τ is retained and used to initialize step $\tau - 1$.

Conversion + Hadamard rotation + MPEF: Both techniques are combined: Hadamard rotation is applied for acti-



(a) Original DiTs (ANN)

(b) Spiking DiTs (Conversion)

(c) Our Spiking DiTs (with MPEF)

Figure 5. Visual comparison of image generation results on DiT-XL-2-256 under $\text{cfg} = 1.5$ and 50 denoising steps. All spiking models set $T = 4, n = 6$.

Table 1. **Quantitative comparison on ImageNet (256×256 & 512×512) under $\text{cfg} = 1.5$.** We compare our method ($T = 4$) with varying thresholds ($n = 6, 8, 10$) against full-precision (FP) baselines and state-of-the-art PTQ methods under W4A8 and W8A8 settings. "Steps" denotes the diffusion steps.

Model	Steps	Method	Low-Bit Setting (W4A8)					High-Bit Setting (W8A8)				
			Config	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec $\uparrow$	Config	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec $\uparrow$
DiT-XL-2-256 ($\text{cfg}=1.5$)	25	FP (Baseline)	W16A16	178.88	5.96	23.79	0.82	W16A16	178.88	5.96	23.79	0.82
		PTQ4DiT	W4A8	66.37	34.33	43.28	0.42	W8A8	129.57	13.03	32.16	0.69
		Q-DiT	W4A8	188.28	8.42	21.68	0.73	W8A8	213.52	7.40	21.50	0.77
		Ours	$T = 4, n = 6$	149.95	6.52	22.01	0.75	$T = 4, n = 8$	175.81	5.99	23.48	0.81
	50	FP (Baseline)	W16A16	182.20	5.75	22.36	0.81	W16A16	182.20	5.75	22.36	0.81
		Q-Diffusion	W4A8	109.22	22.89	29.49	0.57	W8A8	153.01	14.61	27.57	0.66
		PTQ4DiT	W4A8	179.95	9.17	24.29	0.70	W8A8	250.68	5.45	19.50	0.78
		Q-DiT	W4A8	209.42	6.54	18.93	0.76	W8A8	231.47	6.16	19.07	0.80
		Ours	$T = 4, n = 6$	160.33	5.86	20.41	0.76	$T = 4, n = 8$	181.38	5.59	21.61	0.82
	100	FP (Baseline)	W16A16	181.03	5.56	21.21	0.82	W16A16	181.03	5.56	21.21	0.82
		Q-Diffusion	W4A8	42.80	54.95	36.13	0.38	W8A8	202.84	7.93	19.46	0.72
		PTQ4DiT	W4A8	190.38	7.75	22.01	0.72	W8A8	277.27	4.73	17.83	0.82
		Q-DiT	W4A8	216.88	6.20	18.48	0.77	W8A8	242.55	5.82	18.57	0.80
		Ours	$T = 4, n = 6$	158.12	5.91	19.64	0.76	$T = 4, n = 8$	182.50	5.46	20.84	0.81
DiT-XL-2-512 ($\text{cfg}=1.5$)	25	FP (Baseline)	W16A16	156.63	6.40	21.92	0.84	W16A16	156.63	6.40	21.92	0.84
		PTQ4DiT	W4A8	36.19	48.58	65.12	0.44	W8A8	95.86	20.84	38.54	0.77
		Q-DiT	W4A8	151.84	11.20	21.09	0.77	W8A8	184.39	7.89	19.76	0.81
		Ours	$T = 4, n = 8$	150.18	6.59	20.18	0.83	$T = 4, n = 10$	150.62	6.39	21.20	0.83
	50	FP (Baseline)	W16A16	159.16	6.01	21.28	0.84	W16A16	159.16	6.01	21.28	0.84
		PTQ4DiT	W4A8	118.32	19.71	52.27	0.73	W8A8	180.74	7.21	17.56	0.79
		Q-DiT	W4A8	160.36	9.68	20.47	0.78	W8A8	200.33	6.76	19.24	0.83
		Ours	$T = 4, n = 8$	155.02	5.91	19.72	0.83	$T = 4, n = 10$	155.55	5.93	20.59	0.82

vation redistribution, and MPEF retains the residual membrane potential from step τ to initialize step $\tau - 1$.

Table 3 summarizes the contributions of each component to the generative performance of Spiking DiTs. The direct conversion baseline suffers from a catastrophic quality collapse, with the FID reaching up to 198.03. Incorporating Hadamard rotation partially alleviates this by redistributing outliers, reducing the FID to 18.47. However, a substantial gap remains, indicating that spatial pre-processing alone cannot resolve the temporal drifts across the denoising trajectory. The addition of the MPEF mechanism yields a significant breakthrough, further driving the FID down to 6.52. By recycling residual membrane potential to stabilize the denoising path, MPEF effectively bridges the spatiotemporal gap, achieving high-fidelity generation that rivals state-of-

the-art quantization methods.

6.4. Operation Compared with ANNs

To evaluate the operation efficiency of our proposed spiking DiTs, we count the number of basic hardware operations and compare them with ANNs. Following the standard neuromorphic computing conventions, we distinguish between Multiply-Accumulate (MAC) operations and Accumulate (AC) operations. These operation counts are converted into energy consumption by applying the cost factors reported in 45 nm CMOS technology (1 MAC ≈ 4.6 pJ, 1 AC ≈ 0.9 pJ) (Horowitz, 2014). As provided in Table 4, our spiking DiTs convert power-intensive MAC operations into energy-efficient AC operations. This significantly reduces overall energy consumption by deploying potential neuromorphic

Table 2. **Quantitative comparison on ImageNet (256×256 & 512×512) under $\text{cfg} = 1.0$.** We compare our method ($T = 4$) with varying thresholds ($n = 6, 8, 10$) against full-precision (FP) baselines and state-of-the-art PTQ methods under W4A8 and W8A8 settings. "Steps" denotes the diffusion steps.

Model	Steps	Method	Low-Bit Setting (W4A8)					High-Bit Setting (W8A8)				
			Config	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec $\uparrow$	Config	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec $\uparrow$
DiT-XL-2-256 ($\text{cfg}=1.0$)	25	FP (Baseline)	W16A16	89.96	11.34	23.94	0.65	W16A16	89.96	11.34	23.90	0.65
		PTQ4DiT	W4A8	27.38	68.62	51.99	0.25	W8A8	51.07	40.96	45.84	0.50
		Q-DiT	W4A8	72.98	26.61	25.46	0.55	W8A8	87.23	21.03	24.81	0.59
		Ours	$T = 4, n = 6$	69.19	17.82	23.21	0.58	$T = 4, n = 8$	81.54	13.41	25.47	0.62
	50	FP (Baseline)	W16A16	90.81	10.99	23.14	0.65	W16A16	90.81	10.99	23.14	0.65
		PTQ4DiT	W4A8	34.57	56.42	40.35	0.28	W8A8	71.08	23.78	30.12	0.59
		Q-DiT	W4A8	76.09	24.61	26.46	0.55	W8A8	90.79	19.90	26.03	0.59
		Ours	$T = 4, n = 6$	65.71	19.29	19.24	0.56	$T = 4, n = 8$	88.03	11.72	22.94	0.65
	100	FP (Baseline)	W16A16	90.81	10.98	22.67	0.65	W16A16	90.81	10.98	22.67	0.65
		PTQ4DiT	W4A8	65.73	28.90	34.56	0.49	W8A8	83.18	17.62	24.99	0.75
		Q-DiT	W4A8	78.28	24.10	26.27	0.56	W8A8	92.66	19.06	26.00	0.59
		Ours	$T = 4, n = 6$	63.78	19.59	19.23	0.56	$T = 4, n = 8$	87.54	11.79	22.93	0.64
DiT-XL-2-512 ($\text{cfg}=1.0$)	25	FP (Baseline)	W16A16	74.87	15.50	22.35	0.74	W16A16	74.87	15.50	22.35	0.74
		PTQ4DiT	W4A8	19.62	73.67	63.05	0.30	W8A8	37.32	54.23	52.83	0.62
		Q-DiT	W4A8	66.42	29.31	24.34	0.65	W8A8	84.40	21.55	22.56	0.71
		Ours	$T = 4, n = 8$	70.60	18.10	22.11	0.73	$T = 4, n = 10$	70.79	17.03	22.30	0.74
	50	FP (Baseline)	W16A16	77.03	14.58	21.87	0.74	W16A16	77.03	14.58	21.87	0.74
		PTQ4DiT	W4A8	48.62	35.82	28.92	0.58	W8A8	67.61	27.21	25.23	0.64
		Q-DiT	W4A8	71.53	25.81	23.93	0.66	W8A8	89.98	18.76	22.15	0.73
		Ours	$T = 4, n = 8$	75.67	15.70	21.72	0.65	$T = 4, n = 10$	73.03	15.90	21.88	0.74

Table 3. Ablation study on DiT-XL-2-256.

Setting	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec $\uparrow$
ANN	178.88	5.96	23.79	0.82
Conversion	4.23	198.03	295.66	0.05
+ Hadamard	91.48	18.47	34.89	0.65
+ MPEF	144.67	7.01	20.22	0.72
+ Both	149.95	6.52	22.01	0.75

hardware.

Table 4. The operations and energy consumption DiT-XL-2-256.

Config	MACs (G)	ACs (G)
ANN	118.69	0.09
T=4, n=6	0.91	482.05
T=4, n=8	0.91	487.24

6.5. Difference from PTQ methods

We compare our method with quantization methods because both spiking and quantization are processes of discretizing activation values. However, we are fundamentally different from quantization. To clarify the difference between our spiking method and PTQ approaches, we emphasize that our core contribution is not simple activation quantization, but the faithful mapping of the DiT denoising process onto the

spiking neural network paradigm. This yields two unique advantages. First, we transform the computational paradigm: SNNs replace MAC with AC operations, enabling ultra-low-power inference on neuromorphic hardware. Second, we exploit temporal error compensation, carrying the residual membrane potential across denoising steps. We adopt the 8-bit activation-quantized DiT in our experiment, as existing methods perform reasonably well at 8-bit but degrade sharply at lower bit-widths such as 6-bit.

7. Conclusion

In this work, we identified and addressed the fundamental spatiotemporal discrepancies that hinder the conversion of Diffusion Transformers to Spiking Neural Networks. By analyzing the activation, we revealed that the interplay between spatial imbalance and temporal drifts poses a formidable challenge to the ANN-to-SNN conversion process. To this end, we proposed a training-free conversion framework with Multi-Threshold neurons to enhance expressive ability, more importantly, inspired by the observation of temporal distribution similarity, we introduce a Membrane Potential Error-Feedback mechanism for dynamic trajectory stabilization. Extensive experiments demonstrate that our framework achieves generative quality comparable to quantization methods, offering a stepping stone for deploying large DiT models on energy-constrained scenarios. We hope this work serves as a stepping stone for deploying large-scale generative models on neuromorphic hardware.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Details of Spiking DiTs

A.1. Spike-Equivalent Transformers.

To establish a rigorous foundation for our Spiking DiT, our framework builds upon the core insights of SpikeZIP-TF (You et al., 2024), which ensures the mathematical equivalence of non-linear operators between SNNs and their ANNs counterparts. Similar to them, we employ the following process to enable Softmax, LayerNorm, and GeLU activation in SNNs:

$$\mathbf{X}[t] = \mathbf{X}[t-1] + x[t], \quad (23)$$

$$\mathbf{O}[t] = \phi(\mathbf{X}[t]), \quad (24)$$

$$o[t] = \mathbf{O}[t] - \mathbf{O}[t-1] \quad (25)$$

where $\mathbf{X}[t]$ is the accumulated input at t time-step; $x[t]$ is the input at t time-step; $\phi(\cdot)$ is the Softmax, LayerNorm and GeLU activation and $o[t]$ is the output at t time step.

We also employ spike-equivalent self-attention as following:

$$A = Q \cdot K = \sum_{t_1=1}^T Q[t] \cdot \sum_{t_2=1}^T K[t] = \sum_{t=1}^T (S_Q[t] \cdot K[t] + S_K[t] \cdot Q[t] - Q[t] \cdot K[t]) \quad (26)$$

where $Q[t], K[t]$ represent the spike trains of query and key at time-step t_1 and t_2 respectively, $S_Q[t], S_K[t]$ represent the accumulated spike trains of query and key.

Spiking adaLN. In DiTs, the Adaptive Layer Normalization (adaLN) layer is a critical component for conditioning. Formally, for a given input $\mathbf{X}$, the operation is defined as:

$$\text{adaLN}(\mathbf{X}, c) = \gamma(c) \cdot \text{LN}(\mathbf{X}) + \beta(c) \quad (27)$$

where $\gamma(c)$ and $\beta(c)$ are scale and shift parameters derived from the conditioning embedding through a linear projection. In our spiking framework, we incorporate a mathematically equivalent implementation of the adaLN module, which is formulated as follows:

$$\text{adaLN}(\mathbf{X}[t], c) = \begin{cases} \gamma(c) \cdot \text{LN}(\mathbf{X}[t]) + \beta(c), & \text{if } t = 1 \\ \gamma(c) \cdot \text{LN}(\mathbf{X}[t]), & \text{if } t > 1 \end{cases} \quad (28)$$

In our implementation, we simplify this by integrating the full bias $\beta(c)$ into the first time step to ensure numerical consistency with the pre-trained ANN DiTs.

A.2. Threshold Selection for Multi-Threshold Neurons.

Specifically, we sample 10 labels and identify the maximum activation values across all denoising steps. The maximum threshold θ is then determined by multiplying these peak activations by 2 and dividing by the total number of simulation time-steps T .

B. Additional Results

Figure 6 illustrates the evolution of activation distributions throughout the denoising time-steps. The distributions of activations between adjacent denoising steps exhibit substantial similarity.

Figure 7 supplements visualization results of our spiking DiTs and the ANN counterpart on ImageNet 256×256.

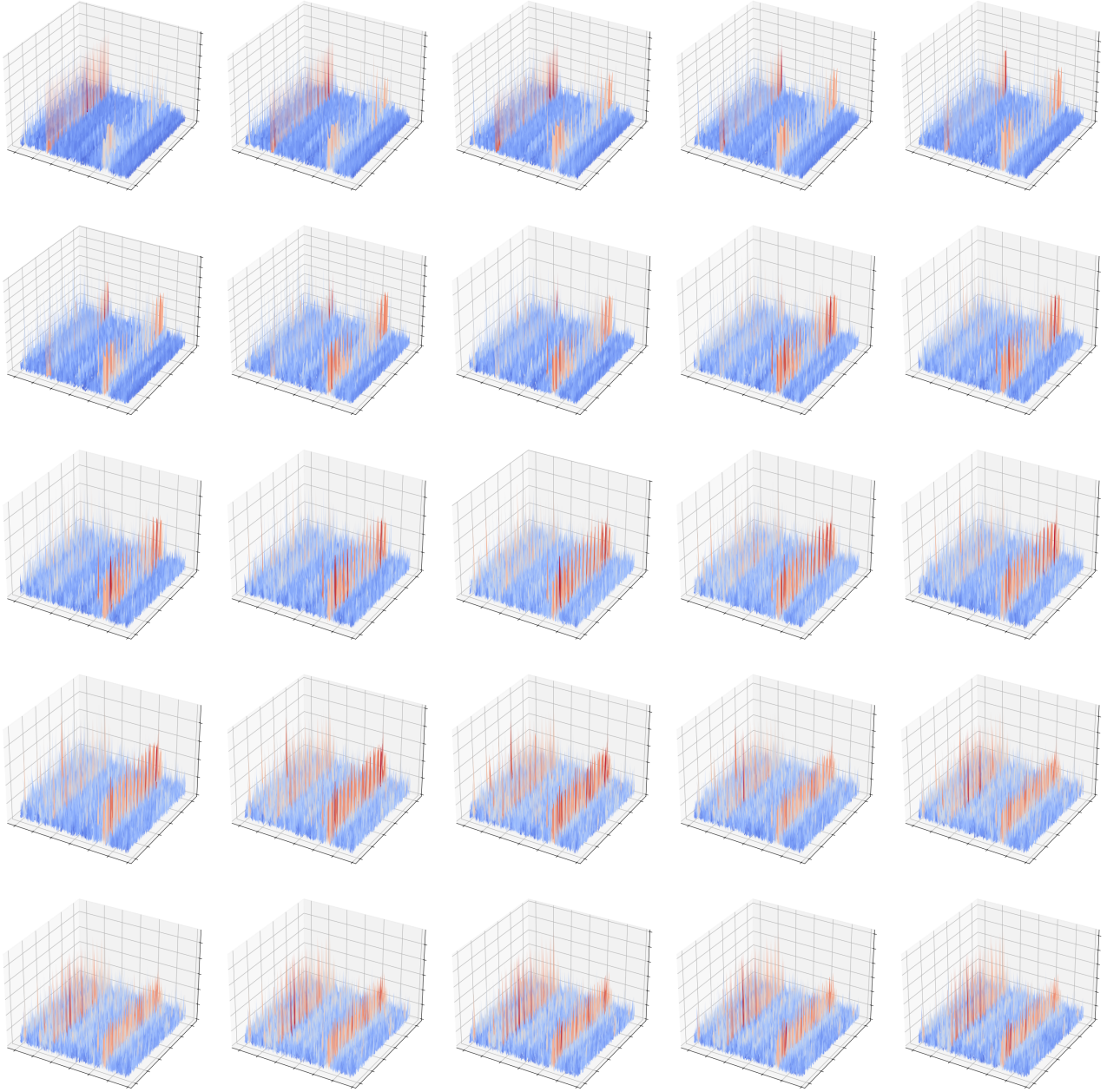
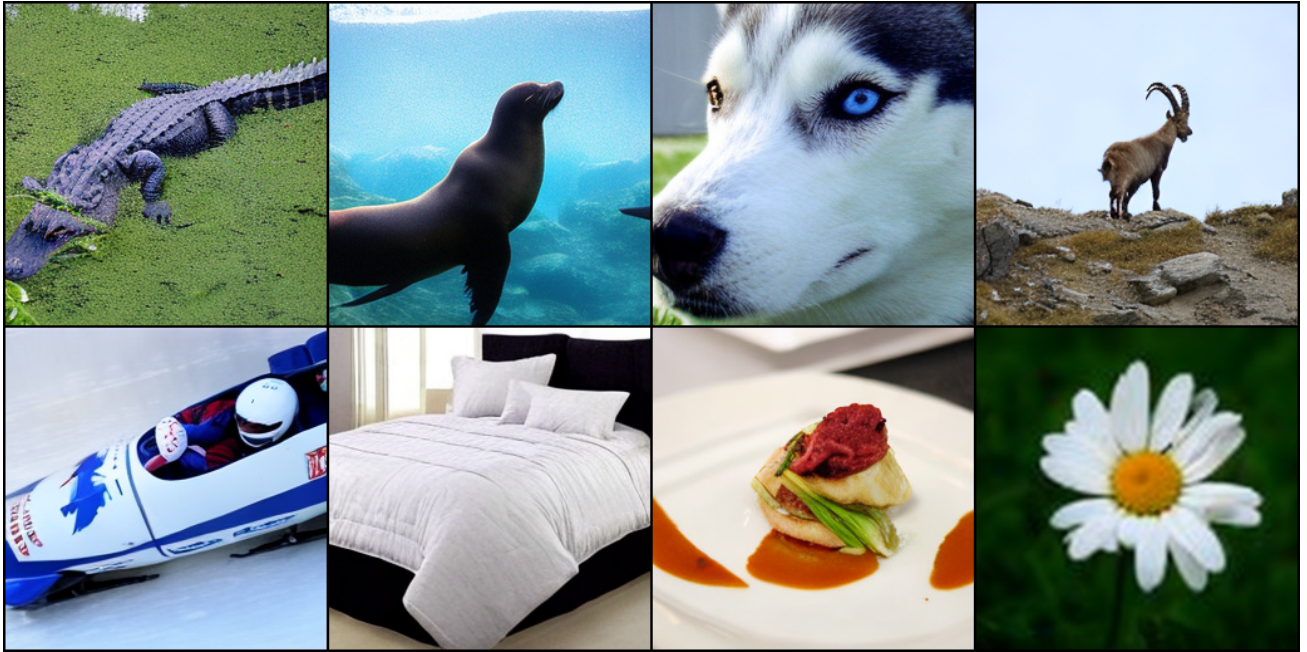


Figure 6. Distributions of activations across the denoising trajectory.



(a) Original DiTs (ANNs)



(b) Spiking DiTs (ours)

Figure 7. Visual comparison of image generation results on DiT-XL-2-256 under $\text{cfg} = 1.5$ and 50 denoising steps. All spiking models set $T = 4$, $n = 6$.